

Inequalities

Worked Solutions with TikZ Number-Line Diagrams

Wisest Maths — Year 1 A-Level, Solving Linear Inequalities (ref ise1)

This document contains all 70 *Solving Linear Inequalities* questions, each with a fully worked solution and a TikZ number-line diagram of the solution set. Open circles denote strict $</>$ and filled circles denote $\leq/\geq$; the shaded ray or segment is the solution. For the *satisfy both* questions the two contributing inequalities are drawn above the axis and their intersection is shown on the axis.

Contents

Solving Linear Inequalities 01 (ise1-001)	1
Solving Linear Inequalities 02 (ise1-002)	1
Solving Linear Inequalities 03 (ise1-003)	2
Solving Linear Inequalities 04 (ise1-004)	2
Solving Linear Inequalities 05 (ise1-005)	3
Solving Linear Inequalities 06 (ise1-006)	3
Solving Linear Inequalities 07 (ise1-007)	4
Solving Linear Inequalities 08 (ise1-008)	4
Solving Linear Inequalities 09 (ise1-009)	4
Solving Linear Inequalities 10 (ise1-010)	5
Solving Linear Inequalities 11 (ise1-011)	6
Solving Linear Inequalities 12 (ise1-012)	6
Solving Linear Inequalities 13 (ise1-013)	7
Solving Linear Inequalities 14 (ise1-014)	7
Solving Linear Inequalities 15 (ise1-015)	8
Solving Linear Inequalities 16 (ise1-016)	8

Solving Linear Inequalities 17 (ise1-017)	9
Solving Linear Inequalities 18 (ise1-018)	9
Solving Linear Inequalities 19 (ise1-019)	10
Solving Linear Inequalities 20 (ise1-020)	10
Solving Linear Inequalities 21 (ise1-021)	11
Solving Linear Inequalities 22 (ise1-022)	11
Solving Linear Inequalities 23 (ise1-023)	12
Solving Linear Inequalities 24 (ise1-024)	12
Solving Linear Inequalities 25 (ise1-025)	13
Solving Linear Inequalities 26 (ise1-026)	13
Solving Linear Inequalities 27 (ise1-027)	14
Solving Linear Inequalities 28 (ise1-028)	14
Solving Linear Inequalities 29 (ise1-029)	15
Solving Linear Inequalities 30 (ise1-030)	16
Solving Linear Inequalities 31 (ise1-031)	16
Solving Linear Inequalities 32 (ise1-032)	17
Solving Linear Inequalities 33 (ise1-033)	17
Solving Linear Inequalities 34 (ise1-034)	18
Solving Linear Inequalities 35 (ise1-035)	18
Solving Linear Inequalities 36 (ise1-036)	19
Solving Linear Inequalities 37 (ise1-037)	19
Solving Linear Inequalities 38 (ise1-038)	20
Solving Linear Inequalities 39 (ise1-039)	20
Solving Linear Inequalities 40 (ise1-040)	21

Solving Linear Inequalities 41 (ise1-041)	22
Solving Linear Inequalities 42 (ise1-042)	22
Solving Linear Inequalities 43 (ise1-043)	23
Solving Linear Inequalities 44 (ise1-044)	23
Solving Linear Inequalities 45 (ise1-045)	24
Solving Linear Inequalities 46 (ise1-046)	25
Solving Linear Inequalities 47 (ise1-047)	25
Solving Linear Inequalities 48 (ise1-048)	25
Solving Linear Inequalities 49 (ise1-049)	26
Solving Linear Inequalities 50 (ise1-050)	27
Solving Linear Inequalities 51 (ise1-051)	27
Solving Linear Inequalities 52 (ise1-052)	28
Solving Linear Inequalities 53 (ise1-053)	28
Solving Linear Inequalities 54 (ise1-054)	29
Solving Linear Inequalities 55 (ise1-055)	29
Solving Linear Inequalities 56 (ise1-056)	30
Solving Linear Inequalities 57 (ise1-057)	30
Solving Linear Inequalities 58 (ise1-058)	31
Solving Linear Inequalities 59 (ise1-059)	31
Solving Linear Inequalities 60 (ise1-060)	32
Solving Linear Inequalities 61 (ise1-061)	33
Solving Linear Inequalities 62 (ise1-062)	33
Solving Linear Inequalities 63 (ise1-063)	34
Solving Linear Inequalities 64 (ise1-064)	35

Solving Linear Inequalities 65 (ise1-065)	35
Solving Linear Inequalities 66 (ise1-066)	36
Solving Linear Inequalities 67 (ise1-067)	36
Solving Linear Inequalities 68 (ise1-068)	37
Solving Linear Inequalities 69 (ise1-069)	38
Solving Linear Inequalities 70 (ise1-070)	38

Solving Linear Inequalities 01 (ise1-001)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $3x - 4 < 11$.

Worked solution.

Step 1: Add 4 to both sides.

$$3x < 15$$

Isolate the term containing x .

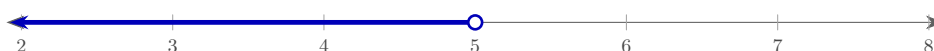
Step 2: Divide both sides by 3.

$$x < 5$$

Dividing by a positive number does not change the direction of the inequality.

Final answer: $x < 5$

Solution on a number line.



Solving Linear Inequalities 02 (ise1-002)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $2x + 5 \geq 13$.

Worked solution.

Step 1: Subtract 5 from both sides.

$$2x \geq 8$$

Isolate the term containing x .

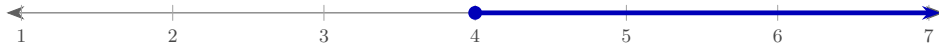
Step 2: Divide both sides by 2.

$$x \geq 4$$

Dividing by a positive number keeps the inequality direction unchanged.

Final answer: $x \geq 4$

Solution on a number line.



Solving Linear Inequalities 03 (ise1-003)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $5x - 3 > 12$.

Worked solution.

Step 1: Add 3 to both sides.

$$5x > 15$$

Isolate the x term.

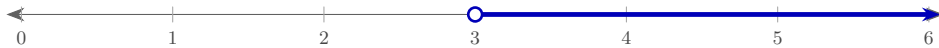
Step 2: Divide both sides by 5.

$$x > 3$$

Positive divisor — direction unchanged.

Final answer: $x > 3$

Solution on a number line.



Solving Linear Inequalities 04 (ise1-004)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $4x + 7 \leq 3$.

Worked solution.

Step 1: Subtract 7 from both sides.

$$4x \leq -4$$

Isolate the x term.

Step 2: Divide both sides by 4.

$$x \leq -1$$

Positive divisor — direction unchanged.

Final answer: $x \leq -1$

Solution on a number line.



Solving Linear Inequalities 05 (ise1-005)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $7 - 2x < 1$.

Worked solution.

Step 1: Subtract 7 from both sides.

$$-2x < -6$$

Bring the constant to the right.

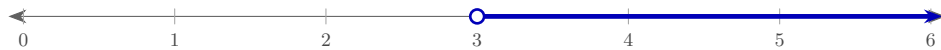
Step 2: Divide both sides by -2 and reverse the inequality.

$$x > 3$$

Dividing by a negative number reverses the direction of the inequality.

Final answer: $x > 3$

Solution on a number line.



Solving Linear Inequalities 06 (ise1-006)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $10 - 3x \geq 1$.

Worked solution.

Step 1: Subtract 10 from both sides.

$$-3x \geq -9$$

Isolate the x term.

Step 2: Divide both sides by -3 and reverse the inequality.

$$x \leq 3$$

Dividing by a negative number flips the inequality.

Final answer: $x \leq 3$

Solution on a number line.



Solving Linear Inequalities 07 (ise1-007)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $8 - 5x > -7$.

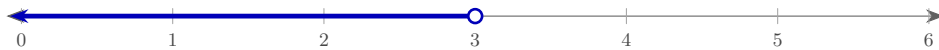
Worked solution.*Step 1: Subtract 8 from both sides.*

$$-5x > -15$$

Isolate the x term.*Step 2: Divide both sides by -5 and reverse the inequality.*

$$x < 3$$

Dividing by a negative reverses the inequality.

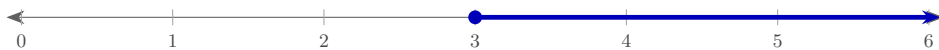
Final answer: $x < 3$
Solution on a number line.**Solving Linear Inequalities 08** (ise1-008)*Difficulty: Foundation / Marks: 2***Find the set of values of x which satisfy $12 - 4x \leq 0$.****Worked solution.***Step 1: Subtract 12 from both sides.*

$$-4x \leq -12$$

Isolate the x term.*Step 2: Divide both sides by -4 and reverse the inequality.*

$$x \geq 3$$

Dividing by a negative flips the inequality.

Final answer: $x \geq 3$
Solution on a number line.**Solving Linear Inequalities 09** (ise1-009)*Difficulty: Foundation / Marks: 2***Find the set of values of x which satisfy $6x + 1 > 4x + 9$.****Worked solution.***Step 1: Subtract $4x$ from both sides.*

$$2x + 1 > 9$$

Collect x terms on the left.

Step 2: Subtract 1 from both sides.

$$2x > 8$$

Isolate the x term.

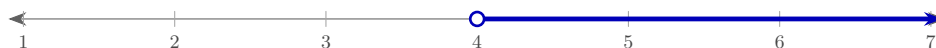
Step 3: Divide both sides by 2.

$$x > 4$$

Positive divisor — direction unchanged.

Final answer: $x > 4$

Solution on a number line.



Solving Linear Inequalities 10 (ise1-010)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $5x - 2 \leq 3x + 8$.

Worked solution.

Step 1: Subtract $3x$ from both sides.

$$2x - 2 \leq 8$$

Collect x terms on the left.

Step 2: Add 2 to both sides.

$$2x \leq 10$$

Isolate x .

Step 3: Divide both sides by 2.

$$x \leq 5$$

Solution found.

Final answer: $x \leq 5$

Solution on a number line.



Solving Linear Inequalities 11 (ise1-011)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $7x + 3 < 4x - 6$.

Worked solution.

Step 1: Subtract $4x$ from both sides.

$$3x + 3 < -6$$

Collect x terms on the left.

Step 2: Subtract 3 from both sides.

$$3x < -9$$

Isolate x .

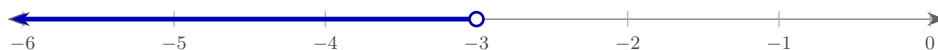
Step 3: Divide both sides by 3.

$$x < -3$$

Positive divisor — direction unchanged.

Final answer: $x < -3$

Solution on a number line.



Solving Linear Inequalities 12 (ise1-012)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $9 - x \geq 7x + 1$.

Worked solution.

Step 1: Add x to both sides.

$$9 \geq 8x + 1$$

Move x to the right to keep its coefficient positive.

Step 2: Subtract 1 from both sides.

$$8 \geq 8x$$

Isolate the x term.

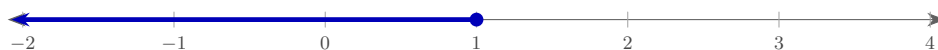
Step 3: Divide both sides by 8.

$$1 \geq x \Rightarrow x \leq 1$$

Rewrite with x on the left.

Final answer: $x \leq 1$

Solution on a number line.



Solving Linear Inequalities 13 (ise1-013)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $4 - 3x < 2x - 11$.

Worked solution.

Step 1: Add $3x$ to both sides.

$$4 < 5x - 11$$

Eliminate the x term from the left.

Step 2: Add 11 to both sides.

$$15 < 5x$$

Isolate x .

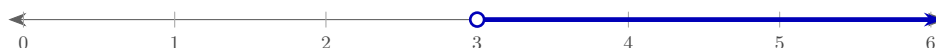
Step 3: Divide both sides by 5.

$$3 < x \Rightarrow x > 3$$

Rewrite with x on the left.

Final answer: $x > 3$

Solution on a number line.



Solving Linear Inequalities 14 (ise1-014)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $8x - 1 \geq 5x + 11$.

Worked solution.

Step 1: Subtract $5x$ from both sides.

$$3x - 1 \geq 11$$

Collect x terms on the left.

Step 2: Add 1 to both sides.

$$3x \geq 12$$

Isolate x .

Step 3: Divide both sides by 3.

$$x \geq 4$$

Solution found.

Final answer: $x \geq 4$

Solution on a number line.



Solving Linear Inequalities 15 (ise1-015)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $3 - 4x > 5 - 2x$.

Worked solution.

Step 1: Add $4x$ to both sides.

$$3 > 5 + 2x$$

Eliminate x from the left.

Step 2: Subtract 5 from both sides.

$$-2 > 2x$$

Isolate x .

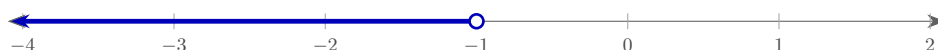
Step 3: Divide both sides by 2.

$$-1 > x \Rightarrow x < -1$$

Rewrite with x on the left.

Final answer: $x < -1$

Solution on a number line.



Solving Linear Inequalities 16 (ise1-016)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $2x - 7 < x + 3$. Represent the solution on a number line.

Worked solution.

Step 1: Subtract x from both sides.

$$x - 7 < 3$$

Collect x terms on the left.

Step 2: Add 7 to both sides.

$$x < 10$$

Isolate x .

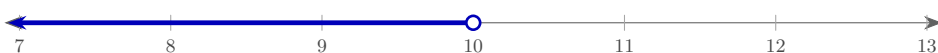
Step 3: On a number line, draw an open circle at 10 and shade to the left.

$$\leftarrow 10$$

Open circle because $x < 10$ (10 is not included).

Final answer: $x < 10$

Solution on a number line.



Solving Linear Inequalities 17 (ise1-017)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $4x + 5 > 2x + 13$. Represent the solution on a number line.

Worked solution.

Step 1: Subtract $2x$ from both sides.

$$2x + 5 > 13$$

Collect x terms on the left.

Step 2: Subtract 5 from both sides.

$$2x > 8$$

Isolate x .

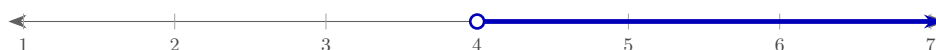
Step 3: Divide both sides by 2. Show open circle at 4, shading right.

$$x > 4$$

Open circle because 4 is not included.

Final answer: $x > 4$

Solution on a number line.



Solving Linear Inequalities 18 (ise1-018)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $6x - 3 \geq 3x + 9$. Represent the solution on a number line.

Worked solution.

Step 1: Subtract $3x$ from both sides.

$$3x - 3 \geq 9$$

Collect x terms.

Step 2: Add 3 to both sides.

$$3x \geq 12$$

Isolate x .

Step 3: Divide both sides by 3. Show closed circle at 4, shading right.

$$x \geq 4$$

Closed circle because $x = 4$ is included.

Final answer: $x \geq 4$

Solution on a number line.



Solving Linear Inequalities 19 (ise1-019)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $5x - 4 \leq 2x + 2$. Represent the solution on a number line.

Worked solution.

Step 1: Subtract $2x$ from both sides.

$$3x - 4 \leq 2$$

Collect x terms.

Step 2: Add 4 to both sides.

$$3x \leq 6$$

Isolate x .

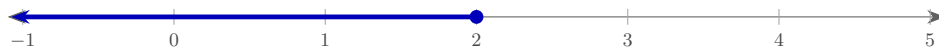
Step 3: Divide both sides by 3. Show closed circle at 2, shading left.

$$x \leq 2$$

Closed circle because $x = 2$ is included.

Final answer: $x \leq 2$

Solution on a number line.



Solving Linear Inequalities 20 (ise1-020)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $11 - 3x < 5 - x$. Represent the solution on a number line.

Worked solution.

Step 1: Add $3x$ to both sides.

$$11 < 5 + 2x$$

Eliminate x from the left.

Step 2: Subtract 5 from both sides.

$$6 < 2x$$

Isolate x .

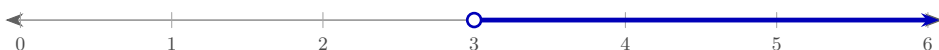
Step 3: Divide both sides by 2. Show open circle at 3, shading right.

$$3 < x \Rightarrow x > 3$$

Open circle at 3.

Final answer: $x > 3$

Solution on a number line.



Solving Linear Inequalities 21 (ise1-021)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $14 - 2x > 6 + 2x$.

Worked solution.

Step 1: Add $2x$ to both sides.

$$14 > 6 + 4x$$

Eliminate x from the left.

Step 2: Subtract 6 from both sides.

$$8 > 4x$$

Isolate x .

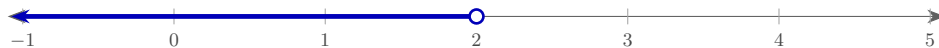
Step 3: Divide by 4.

$$2 > x \Rightarrow x < 2$$

Positive divisor — direction unchanged.

Final answer: $x < 2$

Solution on a number line.



Solving Linear Inequalities 22 (ise1-022)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $9x - 4 < 6x + 2$.

Worked solution.

Step 1: Subtract $6x$ from both sides.

$$3x - 4 < 2$$

Collect x terms on the left.

Step 2: Add 4 to both sides.

$$3x < 6$$

Isolate x .

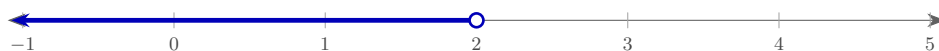
Step 3: Divide by 3.

$$x < 2$$

Solution found.

Final answer: $x < 2$

Solution on a number line.



Solving Linear Inequalities 23 (ise1-023)

Difficulty: Foundation / Marks: 2

Find the set of values of x which satisfy $7x + 2 \leq 4x - 7$.

Worked solution.

Step 1: Subtract $4x$ from both sides.

$$3x + 2 \leq -7$$

Collect x terms.

Step 2: Subtract 2 from both sides.

$$3x \leq -9$$

Isolate x .

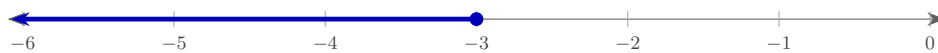
Step 3: Divide by 3.

$$x \leq -3$$

Positive divisor — direction unchanged.

Final answer: $x \leq -3$

Solution on a number line.



Solving Linear Inequalities 24 (ise1-024)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $12x - 9 \leq 4x + 11$. Give your answer in set notation.

Worked solution.

Step 1: Subtract $4x$ from both sides.

$$8x - 9 \leq 11$$

Collect x terms.

Step 2: Add 9 to both sides.

$$8x \leq 20$$

Isolate x .

Step 3: Divide by 8.

$$x \leq \frac{5}{2}$$

$$20 \div 8 = \frac{5}{2}.$$

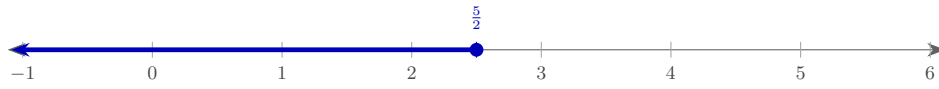
Step 4: Write in set notation.

$$\{x : x \leq \frac{5}{2}\}$$

Use curly brackets and a colon for set notation.

Final answer: $\{x : x \leq \frac{5}{2}\}$

Solution on a number line.



Solving Linear Inequalities 25 (ise1-025)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $5x + 8 > 3 - 2x$. Give your answer in set notation.

Worked solution.

Step 1: Add $2x$ to both sides.

$$7x + 8 > 3$$

Collect x terms.

Step 2: Subtract 8 from both sides.

$$7x > -5$$

Isolate x .

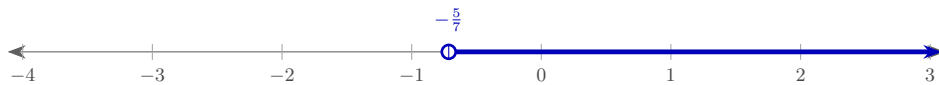
Step 3: Divide by 7 and write in set notation.

$$x > -\frac{5}{7} \Rightarrow \{x : x > -\frac{5}{7}\}$$

Positive divisor — direction unchanged.

Final answer: $\{x : x > -\frac{5}{7}\}$

Solution on a number line.



Solving Linear Inequalities 26 (ise1-026)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $3(x + 4) > 2(x + 7)$.

Worked solution.

Step 1: Expand both brackets.

$$3x + 12 > 2x + 14$$

$$3 \times (x + 4) = 3x + 12 \text{ and } 2 \times (x + 7) = 2x + 14.$$

Step 2: Subtract $2x$ from both sides.

$$x + 12 > 14$$

Collect x terms.

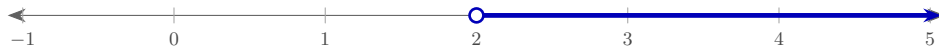
Step 3: Subtract 12 from both sides.

$$x > 2$$

Solution found.

Final answer: $x > 2$

Solution on a number line.



Solving Linear Inequalities 27 (ise1-027)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $4(x - 1) \leq 3(x + 2)$.

Worked solution.

Step 1: Expand both brackets.

$$4x - 4 \leq 3x + 6$$

$$4(x - 1) = 4x - 4 \text{ and } 3(x + 2) = 3x + 6.$$

Step 2: Subtract $3x$ from both sides.

$$x - 4 \leq 6$$

Collect x terms.

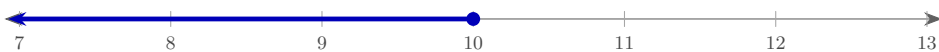
Step 3: Add 4 to both sides.

$$x \leq 10$$

Solution found.

Final answer: $x \leq 10$

Solution on a number line.



Solving Linear Inequalities 28 (ise1-028)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $5(2x - 3) < 3(3x + 1)$. Give your answer in set notation.

Worked solution.

Step 1: Expand both brackets.

$$10x - 15 < 9x + 3$$

$$5(2x - 3) = 10x - 15 \text{ and } 3(3x + 1) = 9x + 3.$$

Step 2: Subtract $9x$ from both sides.

$$x - 15 < 3$$

Collect x terms.

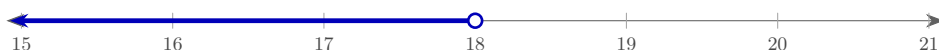
Step 3: Add 15 to both sides and write in set notation.

$$x < 18 \Rightarrow \{x : x < 18\}$$

Solution expressed in set notation.

Final answer: $\{x : x < 18\}$

Solution on a number line.



Solving Linear Inequalities 29 (ise1-029)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $2(5 - x) \geq 3(x - 1)$.

Worked solution.

Step 1: Expand both brackets.

$$10 - 2x \geq 3x - 3$$

$$2(5 - x) = 10 - 2x \text{ and } 3(x - 1) = 3x - 3.$$

Step 2: Add $2x$ to both sides.

$$10 \geq 5x - 3$$

Collect x terms on the right.

Step 3: Add 3 to both sides.

$$13 \geq 5x$$

Isolate x .

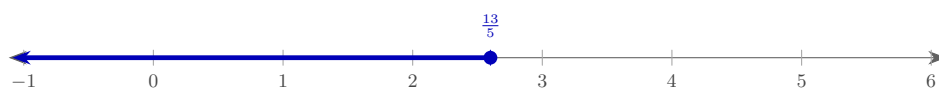
Step 4: Divide by 5.

$$\frac{13}{5} \geq x \Rightarrow x \leq \frac{13}{5}$$

Positive divisor — direction unchanged.

Final answer: $x \leq \frac{13}{5}$

Solution on a number line.



Solving Linear Inequalities 30 (ise1-030)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $6(x + 2) > 4(x + 5)$.

Worked solution.

Step 1: Expand both brackets.

$$6x + 12 > 4x + 20$$

$$6(x + 2) = 6x + 12 \text{ and } 4(x + 5) = 4x + 20.$$

Step 2: Subtract $4x$ from both sides.

$$2x + 12 > 20$$

Collect x terms.

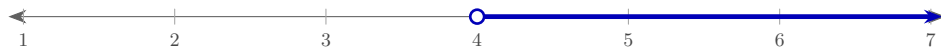
Step 3: Subtract 12 from both sides, then divide by 2.

$$2x > 8 \Rightarrow x > 4$$

Positive divisor — direction unchanged.

Final answer: $x > 4$

Solution on a number line.



Solving Linear Inequalities 31 (ise1-031)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $7(x - 3) \leq 5(x - 1)$. Give your answer in set notation.

Worked solution.

Step 1: Expand both brackets.

$$7x - 21 \leq 5x - 5$$

$$7(x - 3) = 7x - 21 \text{ and } 5(x - 1) = 5x - 5.$$

Step 2: Subtract $5x$ from both sides.

$$2x - 21 \leq -5$$

Collect x terms.

Step 3: Add 21 to both sides.

$$2x \leq 16$$

Isolate x .

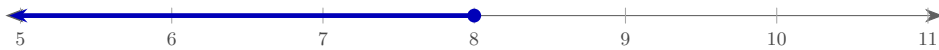
Step 4: Divide by 2 and write in set notation.

$$x \leq 8 \Rightarrow \{x : x \leq 8\}$$

Solution expressed in set notation.

Final answer: $\{x : x \leq 8\}$

Solution on a number line.



Solving Linear Inequalities 32 (ise1-032)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $3(4 - x) < 2(6 - x)$.

Worked solution.

Step 1: Expand both brackets.

$$12 - 3x < 12 - 2x$$

$$3(4 - x) = 12 - 3x \text{ and } 2(6 - x) = 12 - 2x.$$

Step 2: Add $3x$ to both sides.

$$12 < 12 + x$$

Collect x terms on the right.

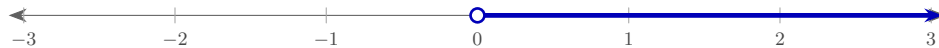
Step 3: Subtract 12 from both sides.

$$0 < x \Rightarrow x > 0$$

Rewrite with x on the left.

Final answer: $x > 0$

Solution on a number line.



Solving Linear Inequalities 33 (ise1-033)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $4(3x - 2) \geq 2(5x + 1)$.

Worked solution.

Step 1: Expand both brackets.

$$12x - 8 \geq 10x + 2$$

$$4(3x - 2) = 12x - 8 \text{ and } 2(5x + 1) = 10x + 2.$$

Step 2: Subtract $10x$ from both sides.

$$2x - 8 \geq 2$$

Collect x terms.

Step 3: Add 8 to both sides, then divide by 2.

$$2x \geq 10 \Rightarrow x \geq 5$$

Positive divisor — direction unchanged.

Final answer: $x \geq 5$

Solution on a number line.



Solving Linear Inequalities 34 (ise1-034)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $9(x - 2) > 5(x + 2)$. Give your answer in set notation.

Worked solution.

Step 1: Expand both brackets.

$$9x - 18 > 5x + 10$$

$$9(x - 2) = 9x - 18 \text{ and } 5(x + 2) = 5x + 10.$$

Step 2: Subtract $5x$ from both sides.

$$4x - 18 > 10$$

Collect x terms.

Step 3: Add 18 to both sides, then divide by 4.

$$4x > 28 \quad \Rightarrow \quad x > 7$$

Positive divisor — direction unchanged.

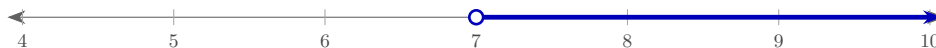
Step 4: Write in set notation.

$$\{x : x > 7\}$$

Solution in set notation.

Final answer: $\{x : x > 7\}$

Solution on a number line.



Solving Linear Inequalities 35 (ise1-035)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $2(7 - 3x) < 4(2 - x)$. Give your answer in set notation.

Worked solution.

Step 1: Expand both brackets.

$$14 - 6x < 8 - 4x$$

$$2(7 - 3x) = 14 - 6x \text{ and } 4(2 - x) = 8 - 4x.$$

Step 2: Add $6x$ to both sides.

$$14 < 8 + 2x$$

Eliminate x from the left.

Step 3: Subtract 8 from both sides.

$$6 < 2x$$

Isolate x .

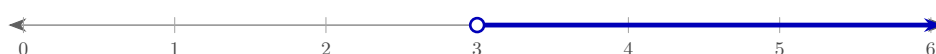
Step 4: Divide by 2 and write in set notation.

$$3 < x \Rightarrow \{x : x > 3\}$$

Rewrite with x on the left.

Final answer: $\{x : x > 3\}$

Solution on a number line.



Solving Linear Inequalities 36 (ise1-036)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $\frac{x}{3} + 2 < 5$.

Worked solution.

Step 1: Subtract 2 from both sides.

$$\frac{x}{3} < 3$$

Isolate the fractional term.

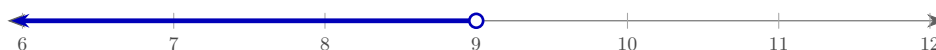
Step 2: Multiply both sides by 3.

$$x < 9$$

Multiplying by a positive number keeps the direction.

Final answer: $x < 9$

Solution on a number line.



Solving Linear Inequalities 37 (ise1-037)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $\frac{2x-1}{4} \geq 3$.

Worked solution.

Step 1: Multiply both sides by 4.

$$2x - 1 \geq 12$$

Clear the denominator. Multiplying by a positive number keeps the direction.

Step 2: Add 1 to both sides.

$$2x \geq 13$$

Isolate x .

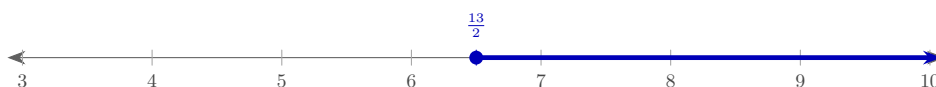
Step 3: Divide by 2.

$$x \geq \frac{13}{2}$$

Solution found.

$$\text{Final answer: } x \geq \frac{13}{2}$$

Solution on a number line.



Solving Linear Inequalities 38 (ise1-038)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $\frac{3x+5}{2} < 7$.

Worked solution.

Step 1: Multiply both sides by 2.

$$3x + 5 < 14$$

Clear the denominator.

Step 2: Subtract 5 from both sides.

$$3x < 9$$

Isolate x .

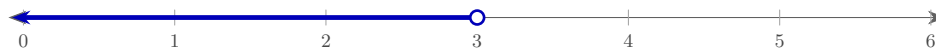
Step 3: Divide by 3.

$$x < 3$$

Solution found.

$$\text{Final answer: } x < 3$$

Solution on a number line.



Solving Linear Inequalities 39 (ise1-039)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $\frac{5-x}{3} > 2$.

Worked solution.*Step 1: Multiply both sides by 3.*

$$5 - x > 6$$

Clear the denominator.

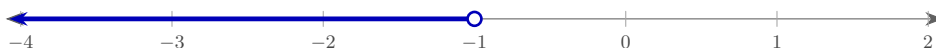
Step 2: Subtract 5 from both sides.

$$-x > 1$$

Isolate x .*Step 3: Multiply both sides by -1 and reverse the inequality.*

$$x < -1$$

Multiplying by a negative reverses the inequality.

Final answer: $x < -1$
Solution on a number line.**Solving Linear Inequalities 40** (ise1-040)*Difficulty: Foundation / Marks: 4*Find the set of values of x which satisfy $\frac{4x-3}{5} \leq x-1$.**Worked solution.***Step 1: Multiply both sides by 5.*

$$4x - 3 \leq 5(x - 1)$$

Clear the denominator.

Step 2: Expand the right side.

$$4x - 3 \leq 5x - 5$$

$$5(x - 1) = 5x - 5.$$

Step 3: Subtract $4x$ from both sides.

$$-3 \leq x - 5$$

Collect x terms.*Step 4: Add 5 to both sides.*

$$2 \leq x \Rightarrow x \geq 2$$

Rewrite with x on the left.
Final answer: $x \geq 2$
Solution on a number line.

Solving Linear Inequalities 41 (ise1-041)

Difficulty: Foundation / Marks: 4

Find the set of values of x which satisfy $\frac{x+3}{2} > \frac{x-1}{3}$. Give your answer in set notation.

Worked solution.

Step 1: Multiply both sides by 6 (LCM of 2 and 3) to clear both denominators.

$$3(x+3) > 2(x-1)$$

$$6 \times \frac{x+3}{2} = 3(x+3) \text{ and } 6 \times \frac{x-1}{3} = 2(x-1).$$

Step 2: Expand both brackets.

$$3x+9 > 2x-2$$

Remove brackets.

Step 3: Subtract $2x$ from both sides.

$$x+9 > -2$$

Collect x terms.

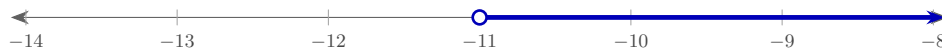
Step 4: Subtract 9 from both sides and write in set notation.

$$x > -11 \Rightarrow \{x : x > -11\}$$

Solution expressed in set notation.

Final answer: $\{x : x > -11\}$

Solution on a number line.



Solving Linear Inequalities 42 (ise1-042)

Difficulty: Foundation / Marks: 4

Find the set of values of x which satisfy $\frac{2x+1}{3} \leq \frac{x+4}{2}$.

Worked solution.

Step 1: Multiply both sides by 6 (LCM of 2 and 3).

$$2(2x+1) \leq 3(x+4)$$

Clears both denominators simultaneously.

Step 2: Expand both brackets.

$$4x+2 \leq 3x+12$$

$$2(2x+1) = 4x+2 \text{ and } 3(x+4) = 3x+12.$$

Step 3: Subtract $3x$ from both sides.

$$x+2 \leq 12$$

Collect x terms.

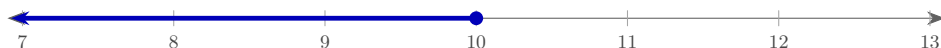
Step 4: Subtract 2 from both sides.

$$x \leq 10$$

Solution found.

Final answer: $x \leq 10$

Solution on a number line.



Solving Linear Inequalities 43 (ise1-043)

Difficulty: Foundation / Marks: 4

Find the set of values of x which satisfy $\frac{6-5x}{2} < \frac{4-8x}{3}$. Give your answer in set notation.

Worked solution.

Step 1: Multiply both sides by 6 (LCM of 2 and 3).

$$3(6-5x) < 2(4-8x)$$

Clears both denominators.

Step 2: Expand both brackets.

$$18-15x < 8-16x$$

$$3(6-5x) = 18-15x \text{ and } 2(4-8x) = 8-16x.$$

Step 3: Add $16x$ to both sides.

$$18+x < 8$$

Collect x terms on the left.

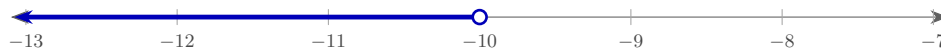
Step 4: Subtract 18 from both sides and write in set notation.

$$x < -10 \Rightarrow \{x : x < -10\}$$

Solution in set notation.

Final answer: $\{x : x < -10\}$

Solution on a number line.



Solving Linear Inequalities 44 (ise1-044)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $\frac{3x-1}{4} \geq 2x$.

Worked solution.

Step 1: Multiply both sides by 4.

$$3x-1 \geq 8x$$

Clear the denominator.

Step 2: Subtract $3x$ from both sides.

$$-1 \geq 5x$$

Collect x terms on the right.

Step 3: Divide by 5.

$$-\frac{1}{5} \geq x \Rightarrow x \leq -\frac{1}{5}$$

Positive divisor — direction unchanged. Rewrite with x on the left.

$$\text{Final answer: } x \leq -\frac{1}{5}$$

Solution on a number line.



Solving Linear Inequalities 45 (ise1-045)

Difficulty: Foundation / Marks: 4

Find the set of values of x which satisfy $\frac{x-2}{2} - \frac{2x+3}{3} < 7$. Give your answer in set notation.

Worked solution.

Step 1: Multiply every term by 6 (LCM of 2 and 3).

$$3(x-2) - 2(2x+3) < 42$$

$$6 \times \frac{x-2}{2} = 3(x-2) \text{ and } 6 \times \frac{2x+3}{3} = 2(2x+3).$$

Step 2: Expand brackets.

$$3x - 6 - 4x - 6 < 42$$

$$3(x-2) = 3x - 6 \text{ and } 2(2x+3) = 4x + 6.$$

Step 3: Simplify the left side.

$$-x - 12 < 42$$

$$3x - 4x = -x \text{ and } -6 - 6 = -12.$$

Step 4: Add 12 to both sides, then divide by -1 and reverse.

$$-x < 54 \Rightarrow x > -54$$

Dividing by -1 reverses the inequality.

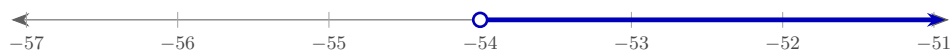
Step 5: Write in set notation.

$$\{x : x > -54\}$$

Solution in set notation.

$$\text{Final answer: } \{x : x > -54\}$$

Solution on a number line.



Solving Linear Inequalities 46 (ise1-046)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-3 < 2x + 1 \leq 9$.

Worked solution.

Step 1: Subtract 1 from all three parts.

$$-4 < 2x \leq 8$$

Treat all three parts simultaneously.

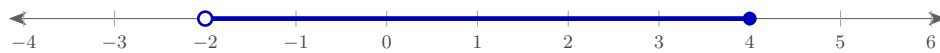
Step 2: Divide all three parts by 2.

$$-2 < x \leq 4$$

Positive divisor — directions unchanged.

Final answer: $-2 < x \leq 4$

Solution on a number line.



Solving Linear Inequalities 47 (ise1-047)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $1 \leq 3x - 5 < 10$.

Worked solution.

Step 1: Add 5 to all three parts.

$$6 \leq 3x < 15$$

Apply the same operation throughout.

Step 2: Divide all three parts by 3.

$$2 \leq x < 5$$

Positive divisor — directions unchanged.

Final answer: $2 \leq x < 5$

Solution on a number line.



Solving Linear Inequalities 48 (ise1-048)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-5 \leq 4 - 3x < 10$. Give your answer in set notation.

Worked solution.

Step 1: Subtract 4 from all three parts.

$$-9 \leq -3x < 6$$

Apply the same operation throughout.

Step 2: Divide all three parts by -3 and reverse both inequality signs.

$$3 \geq x > -2$$

Dividing by a negative reverses all inequalities.

Step 3: Rewrite in conventional order and express in set notation.

$$-2 < x \leq 3 \Rightarrow \{x : -2 < x \leq 3\}$$

Write smaller value on the left.

Final answer: $\{x : -2 < x \leq 3\}$

Solution on a number line.

**Solving Linear Inequalities 49** (ise1-049)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-1 < 5x + 4 < 19$.

Worked solution.

Step 1: Subtract 4 from all three parts.

$$-5 < 5x < 15$$

Apply to all parts.

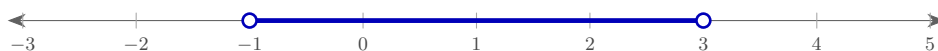
Step 2: Divide all three parts by 5.

$$-1 < x < 3$$

Positive divisor — directions unchanged.

Final answer: $-1 < x < 3$

Solution on a number line.



Solving Linear Inequalities 50 (ise1-050)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $7 \leq 3x - 2 < 16$. Give your answer in set notation.

Worked solution.

Step 1: Add 2 to all three parts.

$$9 \leq 3x < 18$$

Apply to all parts.

Step 2: Divide all three parts by 3 and write in set notation.

$$3 \leq x < 6 \Rightarrow \{x : 3 \leq x < 6\}$$

Positive divisor — directions unchanged.

Final answer: $\{x : 3 \leq x < 6\}$

Solution on a number line.



Solving Linear Inequalities 51 (ise1-051)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-7 < 2 - 3x \leq 14$.

Worked solution.

Step 1: Subtract 2 from all three parts.

$$-9 < -3x \leq 12$$

Apply the same operation.

Step 2: Divide by -3 and reverse both inequalities.

$$3 > x \geq -4$$

Dividing by a negative reverses all inequality signs.

Step 3: Rewrite in conventional order.

$$-4 \leq x < 3$$

Write smaller value on the left.

Final answer: $-4 \leq x < 3$

Solution on a number line.



Solving Linear Inequalities 52 (ise1-052)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $5 \leq 7 + 6x \leq 11$. Give your answer in set notation.

Worked solution.

Step 1: Subtract 7 from all three parts.

$$-2 \leq 6x \leq 4$$

Apply the same operation.

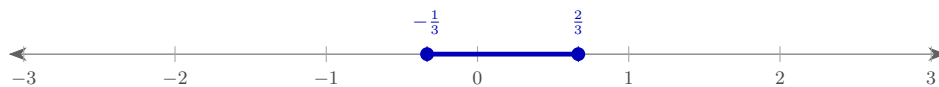
Step 2: Divide all three parts by 6 and write in set notation.

$$-\frac{1}{3} \leq x \leq \frac{2}{3} \Rightarrow \{x : -\frac{1}{3} \leq x \leq \frac{2}{3}\}$$

Positive divisor — directions unchanged.

Final answer: $\{x : -\frac{1}{3} \leq x \leq \frac{2}{3}\}$

Solution on a number line.



Solving Linear Inequalities 53 (ise1-053)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-8 \leq 4 - 2x < 6$.

Worked solution.

Step 1: Subtract 4 from all three parts.

$$-12 \leq -2x < 2$$

Apply the same operation.

Step 2: Divide by -2 and reverse both inequality signs.

$$6 \geq x > -1$$

Dividing by a negative reverses all inequalities.

Step 3: Rewrite in conventional order.

$$-1 < x \leq 6$$

Write smaller value on the left.

Final answer: $-1 < x \leq 6$

Solution on a number line.



Solving Linear Inequalities 54 (ise1-054)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-5 < 2x - 3 < 15$. Give your answer in set notation.

Worked solution.

Step 1: Add 3 to all three parts.

$$-2 < 2x < 18$$

Apply the same operation.

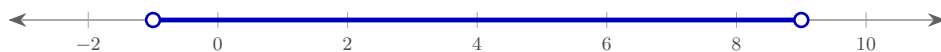
Step 2: Divide all three parts by 2 and write in set notation.

$$-1 < x < 9 \Rightarrow \{x : -1 < x < 9\}$$

Positive divisor — directions unchanged.

Final answer: $\{x : -1 < x < 9\}$

Solution on a number line.



Solving Linear Inequalities 55 (ise1-055)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-4 < \frac{x+1}{2} \leq 3$.

Worked solution.

Step 1: Multiply all three parts by 2.

$$-8 < x + 1 \leq 6$$

Clear the denominator throughout.

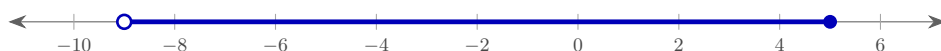
Step 2: Subtract 1 from all three parts.

$$-9 < x \leq 5$$

Isolate x .

Final answer: $-9 < x \leq 5$

Solution on a number line.



Solving Linear Inequalities 56 (ise1-056)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $0 \leq \frac{3x-6}{3} < 4$. Give your answer in set notation.

Worked solution.

Step 1: Multiply all three parts by 3.

$$0 \leq 3x - 6 < 12$$

Clear the denominator.

Step 2: Add 6 to all three parts.

$$6 \leq 3x < 18$$

Isolate the x term.

Step 3: Divide all three parts by 3 and write in set notation.

$$2 \leq x < 6 \Rightarrow \{x : 2 \leq x < 6\}$$

Positive divisor — directions unchanged.

Final answer: $\{x : 2 \leq x < 6\}$

Solution on a number line.



Solving Linear Inequalities 57 (ise1-057)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-2 \leq 5 - 4x < 13$.

Worked solution.

Step 1: Subtract 5 from all three parts.

$$-7 \leq -4x < 8$$

Apply the same operation.

Step 2: Divide by -4 and reverse both inequalities.

$$\frac{7}{4} \geq x > -2$$

Dividing by a negative reverses all inequality signs.

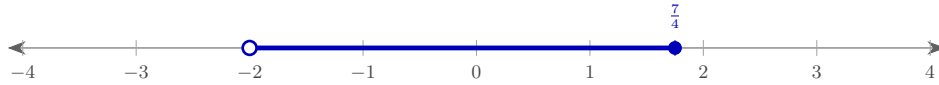
Step 3: Rewrite in conventional order.

$$-2 < x \leq \frac{7}{4}$$

Write smaller value on the left.

Final answer: $-2 < x \leq \frac{7}{4}$

Solution on a number line.



Solving Linear Inequalities 58 (ise1-058)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-3 < 8 + 2x \leq 14$. Give your answer in interval notation.

Worked solution.

Step 1: Subtract 8 from all three parts.

$$-11 < 2x \leq 6$$

Apply the same operation.

Step 2: Divide all three parts by 2.

$$-\frac{11}{2} < x \leq 3$$

Positive divisor — directions unchanged.

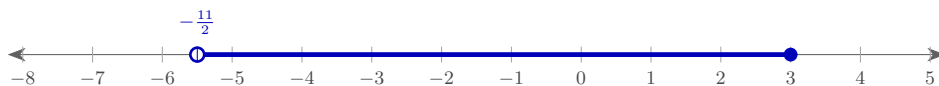
Step 3: Write in interval notation.

$$\left(-\frac{11}{2}, 3\right]$$

Round bracket means $-\frac{11}{2}$ is excluded; square bracket means 3 is included.

Final answer: $\left(-\frac{11}{2}, 3\right]$

Solution on a number line.



Solving Linear Inequalities 59 (ise1-059)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-10 \leq 3x + 2 \leq 17$. Give your answer in interval notation.

Worked solution.

Step 1: Subtract 2 from all three parts.

$$-12 \leq 3x \leq 15$$

Apply the same operation.

Step 2: Divide all three parts by 3.

$$-4 \leq x \leq 5$$

Positive divisor — directions unchanged.

Step 3: Write in interval notation.

$$[-4, 5]$$

Square brackets on both sides since both endpoints are included.

Final answer: $[-4, 5]$

Solution on a number line.



Solving Linear Inequalities 60 (ise1-060)

Difficulty: Foundation / Marks: 3

Find the set of values of x which satisfy $-5 \leq 4 - 3x < 19$. Give your answer in set notation.

Worked solution.

Step 1: Subtract 4 from all three parts.

$$-9 \leq -3x < 15$$

Apply the same operation.

Step 2: Divide by -3 and reverse both inequality signs.

$$3 \geq x > -5$$

Dividing by a negative reverses all inequalities.

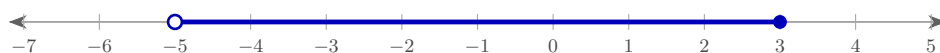
Step 3: Rewrite in conventional order and express in set notation.

$$-5 < x \leq 3 \Rightarrow \{x : -5 < x \leq 3\}$$

Write smaller value on the left.

Final answer: $\{x : -5 < x \leq 3\}$

Solution on a number line.



Solving Linear Inequalities 61 (ise1-061)

Difficulty: Foundation / Marks: 4

Find the set of values of x which satisfy both $3x - 1 < x + 7$ and $2x > x - 2$. Give your answer in set notation.

Worked solution.

Step 1: Solve the first inequality: $3x - 1 < x + 7$.

$$2x < 8 \Rightarrow x < 4$$

Subtract x , add 1, divide by 2.

Step 2: Solve the second inequality: $2x > x - 2$.

$$x > -2$$

Subtract x from both sides.

Step 3: Combine: both conditions must hold simultaneously.

$$-2 < x < 4$$

The solution is the overlap of $x > -2$ and $x < 4$.

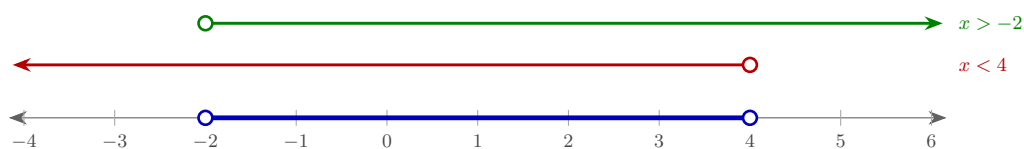
Step 4: Write in set notation.

$$\{x : -2 < x < 4\}$$

Both endpoints are strict (open) inequalities.

Final answer: $\{x : -2 < x < 4\}$

Solution on a number line.



Solving Linear Inequalities 62 (ise1-062)

Difficulty: Foundation / Marks: 4

Find the set of values of x which satisfy both $5x - 2 \leq 3x + 8$ and $4x - 1 > x + 5$. Give your answer in set notation.

Worked solution.

Step 1: Solve $5x - 2 \leq 3x + 8$.

$$2x \leq 10 \Rightarrow x \leq 5$$

Subtract $3x$, add 2, divide by 2.

Step 2: Solve $4x - 1 > x + 5$.

$$3x > 6 \Rightarrow x > 2$$

Subtract x , add 1, divide by 3.

Step 3: Combine both conditions.

$$2 < x \leq 5$$

Overlap of $x > 2$ and $x \leq 5$.

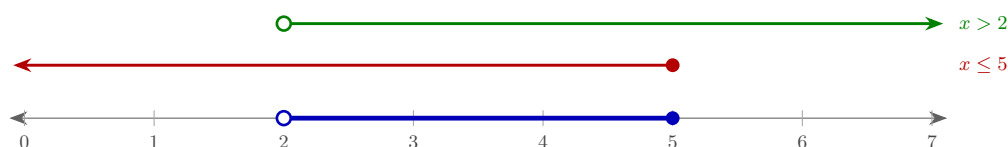
Step 4: Write in set notation.

$$\{x : 2 < x \leq 5\}$$

Open bracket at 2 (strict), closed at 5 (inclusive).

Final answer: $\{x : 2 < x \leq 5\}$

Solution on a number line.



Solving Linear Inequalities 63 (ise1-063)

Difficulty: Foundation / Marks: 4

Find the values of x which satisfy both $2x \geq 3x - 5$ and $3x - 2 \geq x - 6$. Give your answer in set notation.

Worked solution.

Step 1: Solve $2x \geq 3x - 5$.

$$5 \geq x \Rightarrow x \leq 5$$

Subtract $2x$ from both sides: $0 \geq x - 5 \Rightarrow 5 \geq x$.

Step 2: Solve $3x - 2 \geq x - 6$.

$$2x \geq -4 \Rightarrow x \geq -2$$

Subtract x , add 2, divide by 2.

Step 3: Combine both conditions.

$$-2 \leq x \leq 5$$

Overlap of $x \geq -2$ and $x \leq 5$.

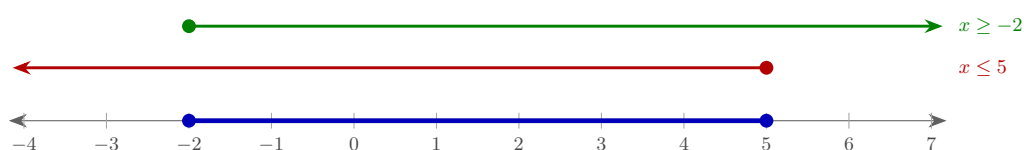
Step 4: Write in set notation.

$$\{x : -2 \leq x \leq 5\}$$

Both endpoints are inclusive.

Final answer: $\{x : -2 \leq x \leq 5\}$

Solution on a number line.



Solving Linear Inequalities 64 (ise1-064)

Difficulty: Foundation / Marks: 4

Find the values of x which satisfy both $4 - 2x < 10$ and $3x - 1 < x + 7$. Give your answer in set notation.

Worked solution.

Step 1: Solve $4 - 2x < 10$.

$$-2x < 6 \Rightarrow x > -3$$

Subtract 4, divide by -2 and reverse.

Step 2: Solve $3x - 1 < x + 7$.

$$2x < 8 \Rightarrow x < 4$$

Subtract x , add 1, divide by 2.

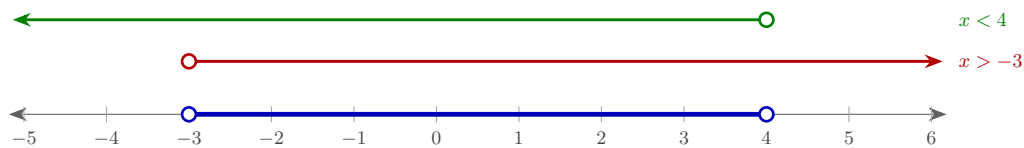
Step 3: Combine both conditions and write in set notation.

$$-3 < x < 4 \Rightarrow \{x : -3 < x < 4\}$$

Both endpoints are strict (open).

Final answer: $\{x : -3 < x < 4\}$

Solution on a number line.



Solving Linear Inequalities 65 (ise1-065)

Difficulty: Foundation / Marks: 4

Find the values of x which satisfy both $5x + 1 \leq 11$ and $2x - 3 < 5x - 6$. Give your answer in set notation.

Worked solution.

Step 1: Solve $5x + 1 \leq 11$.

$$5x \leq 10 \Rightarrow x \leq 2$$

Subtract 1, divide by 5.

Step 2: Solve $2x - 3 < 5x - 6$.

$$3 < 3x \Rightarrow x > 1$$

Subtract $2x$, add 6, divide by 3.

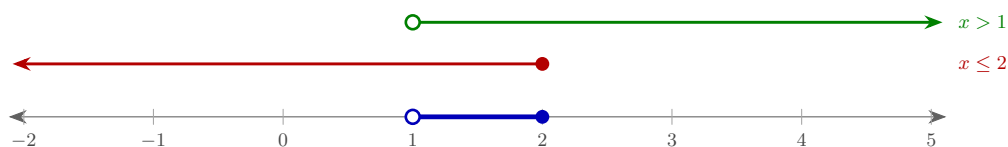
Step 3: Combine both conditions and write in set notation.

$$1 < x \leq 2 \Rightarrow \{x : 1 < x \leq 2\}$$

Overlap of $x > 1$ and $x \leq 2$.

Final answer: $\{x : 1 < x \leq 2\}$

Solution on a number line.



Solving Linear Inequalities 66 (ise1-066)

Difficulty: Foundation / Marks: 4

Find the values of x which satisfy both $2x - 1 \leq 3x - 5$ and $5x - 6 > x + 22$. Give your answer in set notation.

Worked solution.

Step 1: Solve $2x - 1 \leq 3x - 5$.

$$4 \leq x \Rightarrow x \geq 4$$

Subtract $2x$, add 5: $4 \leq x$.

Step 2: Solve $5x - 6 > x + 22$.

$$4x > 28 \Rightarrow x > 7$$

Subtract x , add 6, divide by 4.

Step 3: Combine both conditions.

$$x > 7$$

Since $x > 7$ is more restrictive than $x \geq 4$, the overlap is $x > 7$.

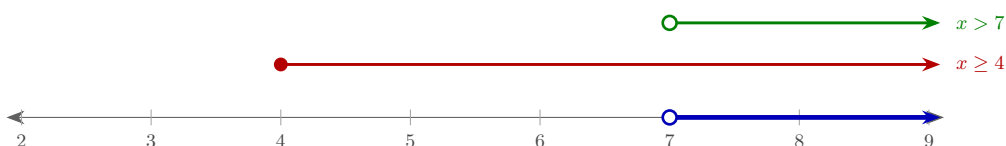
Step 4: Write in set notation.

$$\{x : x > 7\}$$

Only the stricter condition applies.

Final answer: $\{x : x > 7\}$

Solution on a number line.



Solving Linear Inequalities 67 (ise1-067)

Difficulty: Foundation / Marks: 4

Find the values of x which satisfy both $3x + 5 < x + 1$ and $6x - 1 \geq 3x + 5$. Give your answer in set notation.

Worked solution.

Step 1: Solve $3x + 5 < x + 1$.

$$2x < -4 \Rightarrow x < -2$$

Subtract x and 5, divide by 2.

Step 2: Solve $6x - 1 \geq 3x + 5$.

$$3x \geq 6 \Rightarrow x \geq 2$$

Subtract $3x$, add 1, divide by 3.

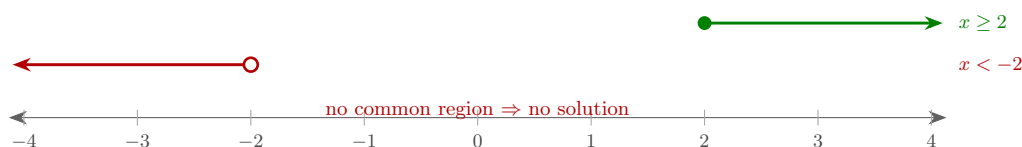
Step 3: Find the overlap: $x < -2$ AND $x \geq 2$.

No overlap - these conditions cannot both be satisfied.

There is no value of x that is simultaneously less than -2 and at least 2.

Final answer: No solution — the two inequalities cannot both be satisfied simultaneously.

Solution on a number line.



Solving Linear Inequalities 68 (ise1-068)

Difficulty: Foundation / Marks: 4

Find the values of x which satisfy both $7x - 4 \geq 3x + 8$ and $5x + 3 \leq 2x + 18$. Give your answer in set notation.

Worked solution.

Step 1: Solve $7x - 4 \geq 3x + 8$.

$$4x \geq 12 \Rightarrow x \geq 3$$

Subtract $3x$, add 4, divide by 4.

Step 2: Solve $5x + 3 \leq 2x + 18$.

$$3x \leq 15 \Rightarrow x \leq 5$$

Subtract $2x$ and 3, divide by 3.

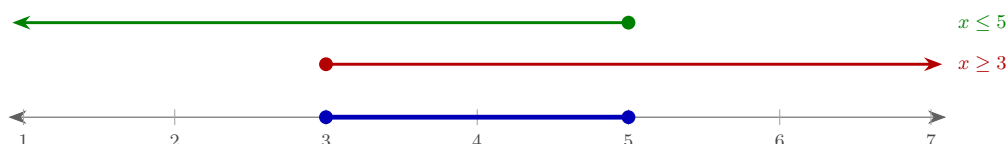
Step 3: Combine both conditions and write in set notation.

$$3 \leq x \leq 5 \Rightarrow \{x : 3 \leq x \leq 5\}$$

Overlap of $x \geq 3$ and $x \leq 5$. Both endpoints inclusive.

Final answer: $\{x : 3 \leq x \leq 5\}$

Solution on a number line.



Solving Linear Inequalities 69 (ise1-069)

Difficulty: Foundation / Marks: 5

Find the values of x which satisfy both $2(x + 3) > x + 8$ and $3(x - 1) \leq 2(x + 4)$. Give your answer in set notation.

Worked solution.

Step 1: Solve $2(x + 3) > x + 8$. Expand: $2x + 6 > x + 8$.

$$x > 2$$

Subtract x and 6 from both sides.

Step 2: Solve $3(x - 1) \leq 2(x + 4)$. Expand: $3x - 3 \leq 2x + 8$.

$$x \leq 11$$

Subtract $2x$, add 3 to both sides.

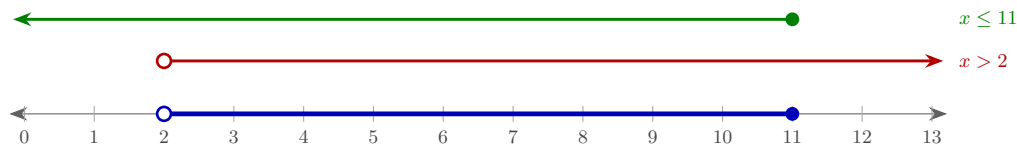
Step 3: Combine both conditions and write in set notation.

$$2 < x \leq 11 \Rightarrow \{x : 2 < x \leq 11\}$$

Overlap of $x > 2$ and $x \leq 11$.

Final answer: $\{x : 2 < x \leq 11\}$

Solution on a number line.



Solving Linear Inequalities 70 (ise1-070)

Difficulty: Foundation / Marks: 5

Find the values of x which satisfy both $4(2x - 1) < 3(3x + 2)$ and $\frac{x + 5}{2} \geq x - 1$. Give your answer in set notation.

Worked solution.

Step 1: Solve $4(2x - 1) < 3(3x + 2)$. Expand: $8x - 4 < 9x + 6$.

$$-4 - 6 < 9x - 8x \Rightarrow -10 < x \Rightarrow x > -10$$

Subtract $8x$ and 6 from both sides.

Step 2: Solve $\frac{x + 5}{2} \geq x - 1$. Multiply both sides by 2: $x + 5 \geq 2x - 2$.

$$5 + 2 \geq 2x - x \Rightarrow 7 \geq x \Rightarrow x \leq 7$$

Subtract x from both sides: $5 \geq x - 2$, then add 2.

Step 3: Combine both conditions and write in set notation.

$$-10 < x \leq 7 \Rightarrow \{x : -10 < x \leq 7\}$$

Overlap of $x > -10$ and $x \leq 7$.

Final answer: $\{x : -10 < x \leq 7\}$

Solution on a number line.

